

100 Days of Azure DevOps

Day 50 Handout — Click-ops is a hobby. IaC is how you sleep: one tool, plan then apply from a pipeline, destroy when the demo ends

5 - Infrastructure as Code · Personal lab only · Educational content · Not a sales pitch

Architecture A — Phase 5 map (Days 41–49)

Day	Object	Keep
41	Declarative vs imperative; drift; one primary tool	ADR in docs/iac-why.md
42–43	ARM JSON literacy: apiVersion, -p @params, outputs	Broccoli you can read when deploys fail
44–45	Bicep resources, modules, what-if	Dress rehearsal before apply
46–48	Terraform verbs, azurerm, backend + lock + modules	State is memory; separate rg-tfstate
49–50	Plan artifact + gated apply in YAML	No laptop-as-prod; destroy after screenshot

Architecture B — mini project bar (not a landing zone)

In	Out	Done when
One RG + storage or a webapp skeleton	Hub-spoke, firewall, AKS	plan+apply from pipeline once
Chosen tool from Day 41	Rewriting the other tool 'just in case'	Recap says why Bicep or why Terraform
Brake on apply	Auto-approve because it is a lab RG you forgot is live	Approval or apply=true demonstrated
Destroy after screenshot	Leaving st* accounts 'for Day 51'	az group list clean or budget-watched leftovers listed

Architecture C — recap failures

Pitfall	What you see	Fix
Both as primary	Two half folders, no green pipeline	Pick one. Literacy notes for the other
Portal still source of truth	File and Azure disagree; you 'fixed' in Portal	Change the file, plan, apply
State leftover	rg-tfstate + blobs after you destroyed workload	Decide: keep for Phase 6 or delete last
Day 42 RGs still billing	Cost recap writes itself	az group delete the orphans

One-liner to remember

Phase 5 recap: click-ops is a hobby; IaC is how you sleep — environments that must survive a weekend live in a file.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal subscription + personal Azure DevOps org. Project azure-100-labs. Screenshot without secrets. Destroy after.

- 1 Provision one RG plus one storage account or a webapp skeleton using the Day 41 primary tool, from the Day 49 pipeline (brake on).
- 2 Save a screenshot of the green plan+apply run (no secrets, no work tenant). Note the deployment name or tf apply ID in docs/phase5-recap.md.
- 3 Write why you picked Bicep or Terraform in that recap. 'Both' is not a reason.
- 4 Destroy once via the same tool (pipeline apply of destroy or terraform destroy / az group delete). Confirm Portal is empty for that RG.

- 5 az group list -o table. Delete leftover rg-day42-lab ... rg-day49-lab if any. rg-tfstate last, only if you are done with it.
- 6 Tomorrow is Docker — do not keep storage accounts as souvenirs.

Definition of done

```
# docs/phase5-recap.md
# Primary: Bicep | Terraform
# Why: _____
# Pipeline run: plan stage URL _____ apply stage URL _____
# Resources: 1 RG + (storage | webapp skeleton)
# Destroy: command/run _____ az group list clean: yes/no
# Leftovers to kill: _____
```

Done checklist

- plan+apply from pipeline once on personal Azure
- Destroy once; leftovers listed or gone
- Written reason for Bicep or Terraform (one)
- Recap stays personal: no employer or client names

Tomorrow — Day 51

Docker fundamentals — images, containers, layer cache. Same app, fewer 'works on my laptop' customs checks.